

Limited Increase in Primary HIV-1C Drug Resistance Mutations in Treatment Naïve Individuals in Ethiopia

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Antiretroviral drug resistance is a major challenge for management and control of HIV-1 infection worldwide and particularly in resource limited countries. The frequency of primary drug resistance mutations (DRMs) and of naturally occurring polymorphisms was determined in 83 antiretroviral treatment (ART) naïve Ethiopian individuals infected with HIV-1, consecutively enrolled in 2010. In all individuals HIV-1C was found. The median (interquartile range) of CD4⁺ T-cell count and viral load were 100 (49–201) cells/ μ l and 44,640 (12,553–134,664) copies/ml, respectively. Protease (PR) and reverse transcriptase (RT) genes of HIV-1 RNA were amplified and sequenced. The proportion of primary DRM to any drug class, using the World Health Organization mutation lists, was 7.2% (6/83), thus exceeding the WHO threshold limit of 5%. Three individuals (3.6%) had non-nucleoside reverse transcriptase inhibitor (NNRTI) mutations, two individuals (2.4%) had protease inhibitor mutations, and one (1.2%) had mutations associated with two drug classes (nucleoside reverse transcriptase inhibitor and NNRTI). In addition, the frequency of polymorphisms in the PR and RT genes was higher compared with previous studies in Ethiopian as well as worldwide isolates. Hence, genotypic drug resistance testing as part of routine management of individuals seems reasonable even in resource limited countries prior to treatment in order to allow proper choice of ART. *J. Med. Virol.* 87:978–984, 2015.

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INTRODUCTION

Infection with human immunodeficiency virus type 1 (HIV-1) and acquired immunodeficiency syndrome (AIDS) remain a major world-wide health problem. In 2012 an estimated 35 million people were living with the virus. Like other sub-Saharan Africa countries, Ethiopia is heavily affected with approximately 760,000 individuals living with HIV/AIDS and an adult prevalence of 1.3% [UNAIDS, 2013]. The availability of antiretroviral treatment (ART) has led to a significant decrease in mortality and morbidity rates of HIV-1 [UNAIDS, 2013]. According to the World Health Organization (WHO) report the number of individuals receiving ART in low- and middle-income nations increased approximately 32-fold from 2002 to 2012 [WHO, 2013]. However, antiretroviral drug resistance remains a major challenge for management and control of HIV-1 infection [Imaz et al., 2011]. The large-scale accessibility of ART regimens in resource-constrained settings also bears the risk of an unnoticed evolution of resistant variants in ART experienced patients and their subsequent transmission [Deeks, 2008]. The awareness of primary DRM led to the recommendation for HIV-1 drug resistance testing as soon as HIV-1 infection is diagnosed [Hirsch et al., 2008]. Primary DRM has declined over time in Europe [Wittkop et al., 2011]. In contrast, in the United States and Africa, the frequency of

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primary DRM is rising [Frentz et al., 2012; Hamers et al., 2013].

In Ethiopia, free ART program was initiated in the late 2005 and since then has been significantly scaled-up with an overall ART coverage of 68% by the end of 2012 (<http://www.etharc.org>, accessed April 08, 2014). There is a dearth of literature on HIV-1 primary DRM, except studies conducted in 2003 [Kassu et al., 2007] and in 2005 [Abegaz et al., 2008] before the introduction of free ART, and a study conducted in 2008 [Mulu et al., 2014] 3 years after commencement of free-of-charge ART distribution to HIV-1 treatment naïve individuals. The former reported between 0 and 2.2% resistance mutations in RT gene. However, increased level of primary DRM reaching 5%, in RT region was reported a few years later [Mulu et al., 2014]. Therefore, the frequency of primary DRMs and the occurrence of polymorphisms was analyzed among treatment naïve Ethiopian individuals infected with HIV-1 enrolled in 2010 in order to assess the trend and potential alteration of primary DRM and polymorphisms.

MATERIALS AND METHODS

Blood samples were collected consecutively in 2010 from 83 HIV-1 infected but ART naïve individuals of Gondar University teaching hospital (n=61), located in the high plateau of northern Ethiopia and of Metema hospital (n=22), located in northwestern Ethiopia on the border to Sudan. The minimum of samples required had been calculated as 49 by considering 95% confidence interval and 3.3% proportion of primary DRM based on the previous study and 5% margin of error. The Institutional Review Board of the University of Gondar approved the study (RPO 55/258/2001).

Blood was drawn from each patient into EDTA treated tubes after a written consent had been obtained. Socio-demographic and clinical data were collected using a pre-designed structured questionnaire. The inclusion criteria for individuals were no history of previous ART, age > 18 years, and no current pregnancy. CD4⁺ T-cell count was performed using Flow Cytometer (FACScount system; Becton Dickinson, San Jose, CA) according to the manufacturer's specifications. HIV was diagnosed based on agglutination and immunochromatographic tests for the detection of antibodies in serum, plasma or whole blood samples using DetermineTM, CapillusTM, and Uni-GoldTM kits (http://pdf.usaid.gov/pdf_docs/PNADO095.pdf).

Samples for HIV viral load determination and nucleic acid sequencing were immediately frozen and were kept below -20°C until used. Extraction, quantitation, amplification, and sequencing of HIV-1 were done as described previously [Mulu et al., 2014]. Briefly, the Abbott m2000sp and m2000rt systems were used for HIV-1 viral load determination. Pro-

tease (PR) and reverse transcriptase (RT) of *pol* encompassing codons 1 to 99 and 1 to 331, respectively, were reverse transcribed and amplified by nested PCR using in-house primers. To demonstrate the absence of contaminating HIV-1 proviral DNA, extracted nucleic acid was also subjected to PCR omitting the RT step, resulting in no amplification product. After gel-purification (Wizard SV Gel and PCR Clean-up kit; Promega, Madison, WI), PCR products were sequenced using the fluorescent di-deoxy-terminator method (Big Dye Terminator kit; Applied Biosystem, Foster City, CA) using an automated population-based sequencer (ABI Prism 310; Applied Biosystem, Foster City, CA). Raw electropherograms were edited manually using Geneious software version 5.4 [Drummond et al., 2011]. FASTA files were analysed according to the list of mutations for surveillance of transmitted HIV-1 drug resistance [Bennett et al., 2009; WHO list 2009]. Subtype determination was done using the Stanford University HIV-1 drug resistance database subtyping tool and was reconfirmed using the Rega HIV-1 subtyping tool, version 2.0 (<http://www.bioafrica.net/subtypetool>, last accession October 14, 2013).

The phylogenetic tree was constructed by the neighbour-joining method from data generated from the *pol* gene region (RT and PR genes), and the reliability of the branching orders was assessed using the bootstrap approach in MEGA version 5 [Tamura et al., 2011]. Nine reference strains representing HIV subtype C were included for comparison. The nucleotide sequences analysed were submitted to GenBank with the accession numbers KJ807649–KJ807731. Samples belonging to transmission clusters were traced back, to exclude possible contamination.

Statistical analysis was carried out using SPSS software, version 18.0 (SPSS, Chicago, IL). Both parametric and non-parametric tests were employed to determine the statistical significance of relationships between DRM of this study and previous reports from Ethiopia, and polymorphism frequencies of present study sequences, reports from Ethiopian isolates and worldwide subtype C isolates. *P*-value < 0.05 was considered to indicate a statistically significant difference. Descriptive statistics were performed for socio-demographic and clinical characteristics. Mean and standard deviation (SD) are presented for normally distributed variables, and median and interquartile range (IQR) for non-normally distributed variables. 95% confidence intervals (CI) were computed by GraphPad Prism software version 5 (San Diego, CA).

RESULTS

Epidemiological, clinical, and virological characteristics of the 83 individuals included in the study are summarized in Table I. Most of them were male (61.4%) and the median (IQR) age was 30 (28–36) years. The median (IQR) of white blood count was 4,600 (3,600–6,000) cells/μl. The differential white

TABLE I. Characteristics of the Study Subjects

Parameters	N	
Gender		
Female	32	38.6%
Male	51	61.4%
Age group in years		
19–28	23	27.7%
29–38	45	54.2%
39–48	10	12.0%
>49	5	6.0%
Baseline CD4 ⁺ T-cells (cells/ μ l)		
<50	22	26.5%
50–99	19	22.9%
100–199	21	25.3%
200–299	17	20.5%
>300	4	4.8%
Baseline white blood count (cells/ μ l)		
1.2–2.99	13	16.7%
3.0–3.99	14	17.9%
4.0–4.99	16	20.5%
>5.0	35	44.9%
not determined	5	–
Baseline hemoglobin (g/dl)		
< 7.5	3	3.9%
7.5–10.5	24	31.2%
10.6–11.9	19	24.7%
>12.0	31	40.3%
not determined	6	–
Baseline HIV RNA load (copies/ml)		
< 2,000	4	4.8%
2,000–10,000	14	16.9%
10,001–100,000	37	44.6%
100,001–500,000	27	32.5%
>500,000	1	1.2%

blood count (mean $\pm$ SD) was neutrophils ($57.5 \pm 13.6\%$) and lymphocytes ($30.8 \pm 10\%$). At time of sampling most of the individuals had opportunistic infections (OIs), e.g., tuberculosis (43.4%), visceral leishmaniasis (6%), oropharyngeal candidiasis (1.2%). The median (IQR) CD4⁺ T-cell count was 100 (49–201) cells/ μ l and the median (IQR) HIV viral load was 44,640 (12,553–134,664) copies/ml. All individuals were infected by HIV-1C, and four monophyletic transmission clusters with bootstrap value of 99% involving eight individuals (9.6%, patient IDs G221/M058, M030/M074, G269/G018, and M029/M039) were detected (Fig. 1). On trace-back possible contamination of corresponding samples was excluded, as samples were collected as well as processed at different time points.

In 6 of the 83 individuals (7.2%) primary DRM was detected (Table II). Eighty three percent of them (5/6) were female, and 67% (4/6) had CD4⁺ T-cell count <100 cells/ μ l and had OIs. Their mean age ($\pm$ SD) was 33.2 ($\pm$ 5.8) years, the median (IQR) viral load was 26,714 (12,528–188,982) copies/ml and the mean ($\pm$ SD) CD4⁺ cell count was 111 ($\pm$ 95.7) cells/ μ l. Distribution of primary DRM had no association with age, or low CD4⁺ T-cell count ($P > 0.05$) but statistically higher primary DRM was found in females compared to males ($P < 0.05$). The median (IQR) viral load of 49,661 (11,382–137,238) and the median (IQR)

CD4⁺ T-cell count of 101 (49–200) of individuals without primary DRM was not statistically different from those with primary DRM.

The frequency of primary DRM to any drug class, based on the WHO mutation lists, was 7.2% (6/83). The prevalence of primary DRM at Gondar and Metema cities, were 6.6% (4/61, CI: 2.2–16.1%) and 9.1% (2/22, CI: 1.3–29%), respectively (Table II). Three (3.6%) individuals had non-nucleoside reverse transcriptase inhibitor (NNRTI) mutations, 2 (2.4%) individuals had protease inhibitor (PI) mutations and one (1.2%) had two-class pDRMs against nucleoside reverse transcriptase inhibitor (NRTI) and NNRTI. The NNRTI associated primary DRMs included: G190A ($n = 2$), and G190E ($n = 1$). Additionally, one patient exhibited two pDRMs, one against NRTI (M184I) and the other against NNRTI (Y181I), and PI mutations (M46I and I85V, respectively) were detected in one patient each.

Polymorphisms were defined as any alteration from subtype B consensus sequence (<http://HIVdb.stanford.edu>, last accession April 04, 2014) and the occurrence of polymorphisms higher than 3.6% was taken into consideration in the present study. Polymorphism frequencies of PR and RT of the present isolates were compared with those of the worldwide subtype C isolates extracted from Stanford HIV database ($n = 7,614$ for PR and $n = 9,150$ for RT). Polymorphism frequencies in the PR were similar to global isolates except position T12 (42.2%) and L19 (49.4%) with lower rates. In the RT however, the polymorphism frequency significantly diverged at more positions. Position K49 (18.1%), A98 (27.7%), K102 (10.8%), D123 (84.3%), I142 (15.7%), A158 (44.6%), K173 (98.8%), Q174 (55.4%), D177 (92.8%), I195 (7.2%), G196 (19.3%), I202 (16.9%), E248 (25.3%), K249 (10.8%), P294 (27.7%), K311 (15.7%), Y319 (4.8%), and E328 (32.5%) with higher rates and S48 (88.1%), V60 (41%), S162 (27.7%), V292 (93%), and I293 (94%) with lower rates (data not shown).

Comparing the polymorphism frequency to Ethiopian HIV-1C infections reported in previous studies [Grossman et al., 2001, data from 1999/2000; Kasso et al., 2007, data from 2003; and Mulu et al., 2014, data from 2008], there is an increasing trend at several positions in PR (I13, G16, N37, L63, T74, and V82) as well as in RT (V60, K102, I202) (Table III).

DISCUSSION

In this study, profiles of drug resistance among treatment naive individuals enrolled in 2010 were investigated. Using in-house primers, amplification and sequencing success rate was 97%. This is higher compared to previous reports conducted in Ethiopia [52%, Abegaz et al., 2008], South Africa [51%, Pillay et al., 2008] and Mozambique [61%, Bartolo et al., 2009].

More than 7% of individuals were found to have primary DRMs based on population genotyping. The

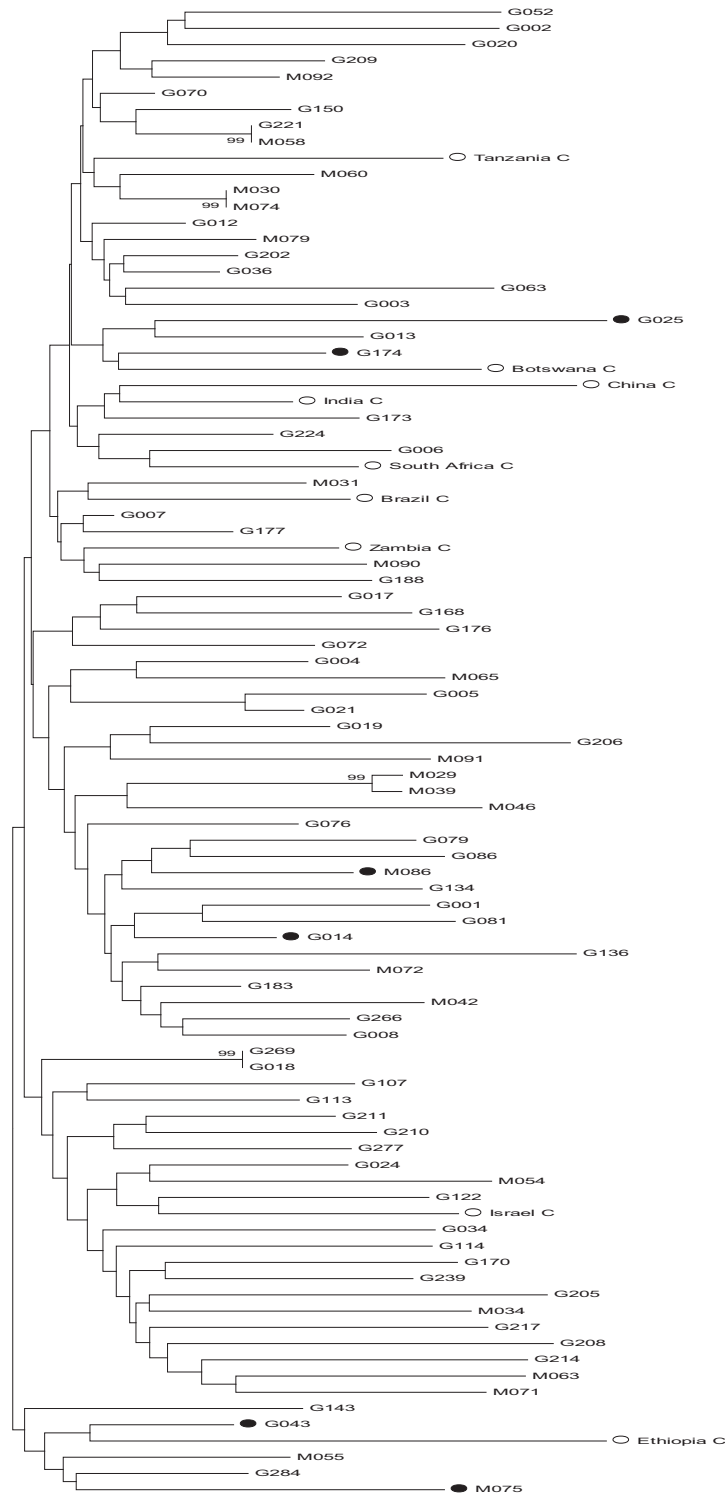


Fig. 1. The neighbour-joining phylogenetic tree depicting the relationship between viral RNA sequences from 83 treatment naïve individuals of Gondar teaching hospital (G) and Metema hospital (M) in northwest Ethiopia. Sequences with primary drug resistance mutations are indicated (black circles). For comparison 9 reference strains (empty circles) from the Los Alamos HIV-1 database representing subtype C with GenBank accession numbers U46016 (Ethiopia C), AF067155 (India C), AF110980 (Botswana C), AF361874 (Tanzania C), AY772699 (S Africa C), U52953 (Brazil C), AF286233 (Israel C), AF286230 (China C), AF286225 (Zambia C) are included. Sequences of this study were submitted to GenBank with the accession numbers KJ807649–KJ807731.

TABLE II. pDRMs in Treatment Naïve Ethiopian Individuals Infected With HIV-1C

Sample ID	Age (years)	Gender	Geographic location	HIV RNA (copies/ml)	CD4 ⁺ T-cells (count/ μ l)	WBC (count/ μ l $\times 10^{-3}$)	Hgb (g/dl)	OIs	pDRMs		
G014	35	Female	Gondar	448,983	253	6.5	10	—	NRTI	NNRTI	PI
G174	30	Female	Gondar	33,222	34	4.5	12.3	—	—	G190A	—
G043	44	Female	Gondar	20,205	62	5.5	17	Opc	—	G190E	—
G025	30	Male	Gondar	102,315	16	1.9	9.3	TB+LD	M184I	Y181I	—
M086	28	Female	Metema	12,916	201	n.d.	n.d.	TB	—	G190A	—
M075	32	Female	Metema	11,363	99	2.9	10	TB	—	—	I85V

WBC, white blood count; Hgb, hemoglobin; OIs, opportunistic infections; pDRMs, primary drug resistance mutations; —, not detected; TB, tuberculosis; LD, visceral leishmaniasis; Opc, oropharyngeal candidiasis; NRTI, nucleoside reverse transcriptase inhibitor; NNRTI, non-NRTI; PI, protease inhibitor.

data of the present study were compared with previous studies conducted in Ethiopian individuals, who reported 2.2% in 2003 [Kassu et al., 2007] and none (0%) in 2005 [Abegaz et al., 2008]. However, the present finding is in accordance with previous reports conducted in sub-Saharan African countries that indicated proportions of primary DRM mutations in treatment naïve individuals ranging from 5% to 12.5% [Ndembu et al., 2008; Tebit et al., 2009; Haidara et al., 2010; Hamers et al., 2012; Mulu et al., 2014]. The results of the present study are in agreement with reports from USA and Europe, which noted reported 12.9% and 10.9% primary DRMs, respectively [Frentz et al., 2012]. Hence, monitoring for primary DRMs in regions of low- and middle-income nations is important to

prevent negative effects of ill-adapted ART [Gupta et al., 2012].

The prevalence of HIV-1 drug resistance in the present study with samples collected 5 years after free access to ART regimes became available in Ethiopia exceeds those reported in studies conducted prior to introduction of ART ($P < 0.05$). Although statistically not different, the primary DRM rate in the present study is higher than the 5% reported from samples obtained 3 years earlier [Mulu et al., 2014]. This suggests there is an increasing trend of DRM in the setting (Fig. 2) and supports the notion that scaling up of ART will drive the emergence of primary DRM. Particularly, unlike recently published findings, major mutations (M46I and I85V) conferring resistance to PIs were detected and this conceivably

TABLE III. Comparative Analysis of the Relative Frequencies of Selected Amino Acid Polymorphisms Detected in HIV-1C Isolates from Treatment Naïve Ethiopian Individuals

	Current study Data from 2010 (n = 83)	Mulu et al. Data from 2008 (n = 160)	Kassu et al. Data from 2003 (n = 92)	Grossmann et al. Data from 1999/2000 (n = 20)
Protease				
L10	3.6	5.0	2.2	0
T12	42.2	45.0	0	0
I13	12.0	5.0	7.0	0
G16	21.7	11.3	16.0	0
L19	49.4	55.0	0	0
K20	14.5	13.8	12.0	25.0
E35	10.8	11.9	6.0	0
M36	96.4	93.0	97.0	95.0
N37	16.9	6.3	4.0	0
L63	65.1	57.5	52.0	20.0
I64	4.8	9.4	1.0	0
T74	15.7	9.4	11.0	0
V82	12.0	7.5	9.0	0
L89	81.9	81.3	93.0	93.0
I93	94.0	88.8	98.0	80.0
Reverse transcriptase				
S48	36.1	31.2	45.0	0
V60	18.1	10.0	14.0	0
A98	27.7	24.4	31.0	20.0
K102	10.8	5.6	3.0	0
I135	26.5	27.5	12.5	0
K166	9.6	15.0	29.2	0
I178	19.3	25.0	8.3	0
I202	16.9	13.8	4.2	0
R211	71.1	53.1	70.8	0

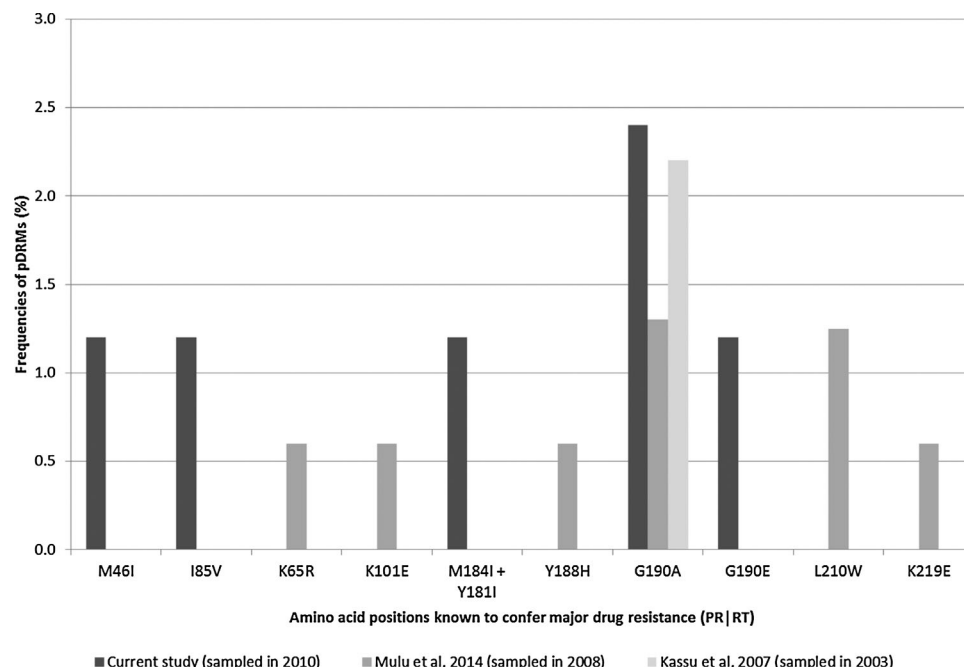


Fig. 2. Comparison of major primary drug resistance mutations identified in treatment naïve Ethiopian individuals infected with HIV-1C. Dark grey column: current study (sampled in 2010), grey column: Mulu et al., 2014 (sampled in 2008), light grey column: Kassu et al., 2007 (samples from 2003).

is a consequence of increased access to PIs since 2008. Currently, however, the I85V mutation has no effect on boosted PIs [Lambert-Niclot et al., 2012].

Three in four primary DRMs identified in this study are related to NNRTI. This is consistent with the current observations of significant increase in NNRTI resistance. While the proportion of NNRTI resistance was below 1% in 2003, it amounted to 6.4% in 2010 [WHO, 2012]. The higher rate of NNRTI related resistance in the present study (4.8%) was also consistent with more recent studies conducted in Ethiopia and other east African countries [Gupta et al., 2012; Mulu et al., 2014]. This may be due to the use of NNRTI as part of standard first-line treatment in combination with the low genetic barrier to resistance. A single nucleotide substitution such as G190A/E may confer a high level of resistance to one substance or even the entire class of NNRTIs [Johnson et al., 2007].

The proportion of polymorphisms varies between and within subtypes, and polymorphisms that are occurring naturally may contribute to the evolution of HIV-1 drug resistance [Svicher et al., 2006; Theys et al., 2013]. In HIV-1B infection, polymorphisms strongly correlate with virological failure and influence polymerase replication capacity [Garriga et al., 2009; Ng et al., 2014]. However, according to some reports the clinical significance of polymorphisms remains controversial [Taylor et al., 2008; Theys et al., 2013].

The comparatively higher abundance of polymorphisms at different positions of the RT gene region in the present study, compared to worldwide HIV-1C isolates, indicates a significant genetic variation of the reverse transcriptase within subtype C [Huang et al., 2012; Theys et al., 2013]. Compared to previous HIV-1 isolates from Ethiopia [Grossmann et al., 2001; Kassu et al., 2007; Mulu et al., 2014] the RT and gene polymorphism rate was higher, which could be due to the dynamic and continuous evolution of the HIV-1C strains over time [Gonzalez et al., 2010], but nevertheless it possibly just reflects the high diversity within subtypes across geographic regions. More longitudinal studies are needed to ascertain the possible association [Huang et al., 2012].

In summary, in the present study primary DRMs are as common as 7.2% and most of them are caused by NNRTI. The proportion of sequences with primary DRM with NRTIs (M184I and Y181I), NNRTIs (G190A/E) and PIs (M46I and I85V) is 1.2%, 3.6% and 2.4%, respectively. The occurrence of PI resistance mutations is reported for the first time in Ethiopia. As the overall frequency of pDRMs has risen beyond the 5% recommended by WHO for threshold surveillance, genotypic drug resistance testing prior to treatment should be considered as part of routine management of individuals even in countries with limited resources in order to choose appropriate first-line ART or at least to allow a rapid switch to alternative drugs.

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